

Enthusiast COURSE

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL : 2022

Test Type : PRACTICE TEST

Test Pattern : NEET (UG)

TEST DATE : _____

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	3	2	2	3	1	1	3	3	4	2	2	4	4	4	3	2	3	2	1	2	1	4	1	4	3	3	1	2	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	4	1	4	4	3	3	3	2	3	2	3	2	3	3	3															



HINT - SHEET

1. **Ans (4)**

$$X_C = \frac{1}{\omega C} = \frac{1}{200 \times 10^{-6}} = 0.5 \times 10^4 = 5 \times 10^3 \Omega$$

$$V_{rms} = \frac{V_0}{\sqrt{2}} = 100$$

$$I_{rms} = \frac{V_{rms}}{X_C} = \frac{100}{5 \times 10^3} = 20 \times 10^{-3} A = 20 \text{ mA}$$

2. **Ans (3)**

It is condition of resonance

$$\therefore V_3 = 220V \text{ and } I = \frac{220}{R} = \frac{220}{200} = 1.1 \text{ A}$$

3. **Ans (2)**

$$\because Z = 0 \text{ } R\sqrt{2} \Rightarrow \cos \phi = \frac{R}{Z} = \frac{R}{R\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \phi = \frac{\pi}{4}$$

4. **Ans (2)**

$$P = i^2 R = \frac{\epsilon^2}{Z^2} R.$$

5. **Ans (3)**

$$\cos \phi = \frac{R}{Z}, Z = \sqrt{R^2 + (\omega L)^2}$$

6. **Ans (1)**

$$P = V \cdot I$$

$$<P> = E_0 \sin \omega t [i_0 (\sin \omega t \cos \phi + \cos \omega t \sin \phi)]$$

$$= \frac{E_0 I_0}{2} \cos \phi$$

7. **Ans (1)**

$$\tan \phi = \frac{V_L - V_C}{V_R} = \frac{100 - 40}{80} = \frac{6}{8} = \frac{3}{4}$$

$$P.f = \cos \phi = \frac{4}{5} = 0.8$$

8. **Ans (3)**

$$\omega = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow L = \frac{1}{\omega^2 C} = \frac{1}{(100)^2 \times 100 \times 10^{-6}} = 1H$$

9. **Ans (3)**

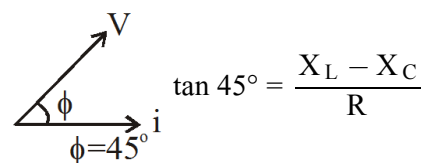
$$\cos \phi = \frac{R}{Z} = \frac{12}{15} = 0.8$$

10. **Ans (4)**

$$i = a \sin \omega t + a \sin (\omega t + 2\pi/3)$$

So peak is a value = 4

11. **Ans (2)**



$$R = 2\pi f L - \frac{1}{2\pi f C} \Rightarrow \frac{1}{2\pi f C} = 2\pi f L - R$$

$$\Rightarrow C = \frac{1}{(2\pi f L - R) 2\pi f}$$

12. **Ans (2)**

$$I = 4 \sin \left(100\pi \times 0 + \frac{\pi}{6} \right) = 4 \times \frac{1}{2} = 2A$$

13. Ans (4)

$$\cos \phi = \frac{R}{Z}$$

$$R = 100 \times \frac{\sqrt{3}}{2} = 50\sqrt{3} \text{ ohm}$$

14. Ans (4)

$$\phi = \pi/2$$

$$\langle P \rangle = V_{\text{rms}} I_{\text{rms}} \cos \pi/2 = 0$$

15. Ans (4)

$$\phi = \pi/2$$

$$\langle P \rangle = 0$$

16. Ans (3)

$$i_{\text{rms}} = \sqrt{\frac{\int_2^4 4t dt}{\int_2^4 dt}} = \sqrt{\frac{4\left(\frac{t^2}{2}\right)_2^4}{(t)_2^4}} = \sqrt{\frac{2[4^2 - 2^2]}{(4-2)}} = 2\sqrt{3} \text{ A}$$

18. Ans (3)

$$\phi = 45^\circ \quad \tan 45^\circ = \frac{X}{R} \Rightarrow X = R$$

$$\begin{aligned} X &= X_L && \Rightarrow RL \\ &= X_C && \Rightarrow RC \\ &= X_L \sim X_C && \Rightarrow LCR \end{aligned}$$

19. Ans (2)

$$\text{at } f = 0 \quad X_L = 0, X_C = \infty \quad Z = \infty \quad I = 0$$

$$\text{at } f = \infty \quad X_L = \infty, X_C = 0 \quad Z = \infty \quad I = 0$$

20. Ans (1)

Potential lags current. So this is RC circuit.

$$\tan \frac{\pi}{4} = \frac{X_C}{R} \Rightarrow X_C = R$$

$$\text{If } C = 10 \mu\text{F} \Rightarrow X_C = \frac{1}{\omega C}$$

$$X_C = \frac{1}{100 \times 10^{-6}} = 1000 \Omega$$

21. Ans (2)

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{R^2 + \left(R - \frac{R}{2}\right)^2}$$

$$= \sqrt{R^2 + \frac{R^2}{4}} = \frac{\sqrt{5}R}{2}$$

$$\tan \phi = \frac{X_L - X_C}{R} = \frac{R - \frac{R}{2}}{R} = \frac{1}{2}$$

$$\phi = \tan^{-1} \frac{1}{2}$$

22. Ans (1)

$$R = 100 \Omega \text{ (given)}$$

when capacitor is removed then $\phi = 45^\circ$

$$\text{so } \tan \phi = \frac{X_L}{R} = 1$$

$$X_L = R$$

when inductor is removed

then $\phi = 45^\circ$

$$\tan \phi = \frac{X_C}{R} = 1$$

$$X_C = R$$

$$\text{so } X_L = X_C = R$$

so circuit is in resonance so $Z = R$

$$I = \frac{V}{R} = \frac{100}{100} = 1 \text{ A}$$

23. Ans (4)

$$\text{For } n_r = n \Rightarrow X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C} \quad \therefore \phi = 0$$

$$\text{If } n_r < n \Rightarrow X_L > X_C \Rightarrow \omega L > \frac{1}{\omega C}$$

$$\therefore \phi = +ve$$

Current lags behind voltage.

25. Ans (4)

$$X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10 \Omega \text{ and}$$

$$X_C = \frac{1}{2000 \times 50 \times 10^{-6}} = 10 \Omega$$

Total impedance of the circuit

$$Z = 6 + \sqrt{(R)^2 + (X_L - X_C)^2}$$

$$= 6 + \sqrt{(4)^2 + 0} = 10 \Omega$$

Ammeter reads r.m.s. current so it's value

$$i_{\text{rms}} = \frac{V_{\text{rms}}}{\text{Total impedance}} = \frac{20/\sqrt{2}}{10}$$

$$= \sqrt{2} = 1.41 \text{ A}$$

Since $X_L = X_C$; this is the condition of

resonance and in this condition $V = V_R$

$$= iR = 1.4 \times 4 = 5.6 \text{ V.}$$

26. Ans (3)

$$P = I_{\text{rms}}^2 R$$

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{200}{100} = 2 \text{ A}$$

$$P = 4 \times 36 = 144 \text{ W}$$

27. **Ans (3)**

$$\begin{aligned}
 V &= \sqrt{V_R^2 + (V_L - V_C)^2} \\
 &= \sqrt{(80)^2 + (100 - 40)^2} = 100V
 \end{aligned}$$

28. **Ans (1)**

$$\begin{aligned}
 X_L &= X_C \Rightarrow Z = R \\
 I &= \frac{V}{Z} = \frac{110}{55} = 2A
 \end{aligned}$$

29. **Ans (2)**

$$\begin{aligned}
 P_{av} &= V_{rms} I_{rms} \cos \theta \\
 &= I_{rms}^2 Z \cos \theta \quad \left[\because \cos \theta = \frac{R}{Z} \right] \\
 P_{av} &= \frac{I_0^2}{2} R
 \end{aligned}$$

30. **Ans (3)**

$$\begin{aligned}
 f &= \frac{\omega}{2\pi} = \frac{20\pi \times 10^6}{2\pi} = 10 \times 10^6 = 10^7 \text{ Hz} \\
 \lambda &= \frac{2\pi}{k} = \frac{2\pi}{\pi/2 \times 10^{-2}} = 400\text{m}
 \end{aligned}$$

31. **Ans (4)**

$$\frac{E_0}{B_0} = c \Rightarrow B_0 = \frac{10^{18}}{3 \times 10^8} = 0.33 \times 10^{10}$$

32. **Ans (1)**

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

33. **Ans (4)**

$$c = \frac{E_0}{B_0} \quad \vec{V} = \vec{E} \times \vec{B} \quad \therefore (\vec{V} \perp \vec{E} \perp \vec{B})$$

38. **Ans (2)**

$$\begin{aligned}
 \vec{P} &= \frac{1}{\mu_0} [\vec{E} \times \vec{B}], \vec{B} = \mu_0 \vec{H} \\
 \vec{P} &= \vec{E} \times \vec{H}
 \end{aligned}$$

45. **Ans (3)**

$$f = \frac{c}{\lambda} \Rightarrow \lambda = \frac{c}{f} = \frac{3 \times 10^8}{40 \times 10^6} = 7.5 \text{ m}$$